

The Effect of Weather on Istanbul's Daily Traffic Index

1. Executive Summary

This project investigates the relationship between meteorological conditions and traffic congestion in Istanbul. By analyzing nearly seven years of data (2018–2024), the study merges traffic indices from the İBB Open Data Portal with weather records from Open-Meteo. Through statistical hypothesis testing and machine learning, the analysis confirms that while the "Day of the Week" is the primary driver of traffic, precipitation and temperature are statistically significant congestion multipliers. The project concludes with a Random Forest predictive model that explains approximately 42% of the variance in daily traffic.

2. Motivation and Objectives

Istanbul has heavy traffic because of its geography and large population. Road capacity is limited, but daily conditions change. Weather is one of the factors that can affect how people travel.

Objective: To measure how much rain, snow, wind, and temperature change the Daily Traffic Index.

Hypothesis (H_1): Bad weather, such as rain and very cold temperatures, causes a statistically significant increase in traffic compared to clear and mild days.

3. Methodology and Data Processing

3.1 Data Sources

The study uses two datasets covering the period from **January 1, 2018 to October 18, 2024**:

1. **Traffic Data:** Daily Average Traffic Index from the İBB Open Data Portal.
2. **Weather Data:** Daily precipitation, rain, snowfall, average temperature, and maximum wind speed from the Open-Meteo API.

3.2 Data Wrangling

The main difficulty was matching the two datasets. The traffic data had non-standard time formats and included double time-zone information. I used Python (pandas) to clean and convert these timestamps, then merged the datasets using an inner join. After cleaning, the final dataset contains 2482 days, which is large enough for reliable statistical analysis.

3.3 Tools Used

- **Python (Pandas, NumPy):** Data cleaning and time-series processing.
- **Matplotlib / Seaborn:** Data visualization.
- **SciPy:** Statistical tests (t-tests).
- **Scikit-Learn:** Machine learning models and evaluation.

4. Exploratory Data Analysis (EDA)

Before building any models, I first explored the data to understand its main patterns.

4.1 The Weekly Pulse

The strongest pattern in the data is the weekly cycle. Boxplots clearly show a large and consistent drop in traffic on weekends.

Insight: On weekdays, the median traffic index is usually between 30 and 35, while on weekends it falls to around 15–20. This shows that the day of the week has a major impact on traffic and must be accounted for when analyzing weather effects.

4.2 Weather Correlations

Precipitation: Shows a positive correlation with traffic (more rain = higher index).

Temperature: Shows a negative correlation. As temperature drops, traffic density rises.

Wind: Shows a very weak, almost flat correlation with traffic.

5. Statistical Findings (Hypothesis Testing)

To go beyond visual patterns, I ran Independent T-tests using a significance level of $\alpha = 0.05$.

5.1 Rain (The Most Common Disruptor)

Test: Compared average traffic on rainy days (> 0 mm) with days without rain.

Result: $p\text{-value} = 1.45 \times 10^{-15}$

Conclusion: Highly significant. Rain consistently increases traffic and is the most reliable weather-related cause of congestion.

5.2 Snow (Rare but Impactful)

Test: Compared average traffic on days with snowfall (> 0 cm) versus days without snow.

Result: $p\text{-value} = 0.042$

Conclusion: Statistically significant. Although snow is uncommon in Istanbul, when it does occur it clearly disrupts traffic, allowing us to reject the null hypothesis.

5.3 Temperature (Cold vs. Warm Days)

Test: Compared days below the median temperature (15.4°C) with days above it.

Result: $p\text{-value} = 2.89 \times 10^{-9}$

Conclusion: Statistically significant. Traffic is consistently heavier on colder days.

5.4 Wind Speed (Little to No Effect)

Test: Compared high-wind days (above the median) with low-wind days.

Result: $p\text{-value} > 0.05$ (not significant)

Conclusion: Not significant. Unlike rain or snow, normal wind levels do not meaningfully affect traffic in Istanbul. Only extreme cases (such as bridge closures), which are rare outliers, appear to have any noticeable impact.

6. Detailed Analysis & Machine Learning

I built a predictive pipeline to estimate the daily traffic index using date and weather variables.

6.1 The Precipitation

The analysis separates two different types of weather effects:

Rain (Routine Friction): Rain occurs on many days and gradually increases traffic. Reduced visibility and lower driving speeds cause a steady rise in congestion.

Snow (Disruptive Shocks): Snow is rare, but when it happens it can seriously reduce road capacity. In some areas, steep roads become unusable, leading to near-paralysis in traffic.

6.2 Model Comparison

To avoid data leakage, I used a time-series split, training on past data and testing on future data.

Model	R^2 Score	RMSE	Interpretation
Linear Regression	0.28	4.82	Weak performance. It failed to capture non-linear effects such as snow shocks and weekend drops.
Random Forest	0.42	3.90	Better results. It handled conditional patterns (for example, heavy rain on a weekday causes major congestion, while the same rain on a Sunday has a smaller effect).

6.3 Feature Importance

The Random Forest model ranked the variables as follows:

- **Day of Week / Weekend:** Primary factor ($\approx 55\%$)
- **Temperature:** Secondary factor ($\approx 20\%$)
- **Rain:** Short-term impact ($\approx 15\%$)
- **Snow:** Rare but strong effect ($\approx 8\%$)
- **Wind Speed:** Negligible ($< 2\%$)

7. Conclusion

This study shows that weather acts as a congestion multiplier in Istanbul.

Rain is a frequent and predictable factor that consistently increases traffic.

Snow is rare but creates sharp and statistically significant traffic spikes.

Temperature sets a seasonal baseline, with colder periods being more congested.

Wind has no meaningful effect on daily traffic levels.

By combining calendar information with rain and snow data, the Random Forest model can explain about 42% of daily traffic variation. This suggests that short-term traffic conditions in Istanbul can be partially predicted using weather forecasts.

References

Data Sources

Istanbul Metropolitan Municipality (İBB). (2024). *Daily Traffic Index Data*. İBB Open Data Portal. Retrieved from <https://data.ibb.gov.tr/>

Open-Meteo. (2024). *Historical Weather Data API*. Retrieved from <https://open-meteo.com/>